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Biogenesis of a specialized plant–fungal interface during host cell internalization of *Golovinomyces orontii* haustoria

Cristina O. Micali, Ulla Neumann, Dorit Grunewald, Ralph Panstruga and Richard O'Connell*
Max-Planck-Institute for Plant Breeding Research, Department of Plant–Microbe Interactions, Carl-von-Linné-Weg 10, 50829 Köln, Germany.

Summary

Powdery mildew fungi are biotrophic pathogens that require living plant cells for their growth and reproduction. Elaboration of a specialized cell called a haustorium is essential for their pathogenesis, providing a portal into host cells for nutrient uptake and delivery of virulence effectors. Haustoria are enveloped by a modified plant plasma membrane, the extrahaustorial membrane (EHM), and an extrahaustorial matrix (EHMx), across which molecular exchange must occur, but the origin and composition of this interfacial zone remains obscure. Here we present a method for isolating *Golovinomyces orontii* haustoria from Arabidopsis leaves and an ultrastructural characterization of the haustorial interface. Haustoria were progressively encased by deposits of plant cell wall polymers, delivered by secretory vesicles and multivesicular bodies (MVBs) that ultimately become entrapped within the encasement. The EHM and EHMx were not labelled by antibodies recognizing eight plant cell wall and plasma membrane antigens. However, plant resistance protein RPW8.2 was specifically recruited to the EHMs of mature haustoria. Fungal cell wall-associated molecular patterns such as chitin and β-1,3-glucans were exposed at the surface of haustoria. Fungal MVBs were abundant in haustoria and putative exosome vesicles were detected in the paramural space and EHMx, suggesting the existence of an exosome-mediated secretion pathway.

Received 15 July, 2010; revised 31 August, 2010; accepted 31 August, 2010. *For correspondence. E-mail oconnel@mpiz-koeln.mpg.de; Tel. (+49) 221 5062 311; Fax (+49) 221 5062 353.

Introduction

Powdery mildew fungi infect over 10 000 plant species worldwide, and cause yield losses of up to 40% on many economically important crops (Takamatsu, 2004; Hückelhoven, 2005; Micali *et al*., 2008). These pathogens are obligate biotrophs that require living plant cells for their growth and reproduction. Four species of powdery mildew, namely *Golovinomyces orontii*, *G. cichoracearum*, *Erysiphe cruciferarum* and *Oidium neolycopersici*, infect the model plant Arabidopsis and are therefore valuable pathosystems for studying the molecular basis of obligate biotrophic parasitism (reviewed in Micali *et al*., 2008).

Most species of powdery mildew fungi exclusively colonize plant epidermal cells by directly penetrating the plant cuticle and cell wall using infection structures called appressoria. Subsequently, the fungal penetration hypha differentiates to form a specialized haustorium inside a living host cell. Powdery mildew haustoria are determinate, unicellular structures composed of a globular central body from which filamentous lobes protrude (Gil and Gay, 1977; O'Connell and Panstruga, 2006). During host cell invasion, haustoria expand and invaginate the host plasma membrane, which loses many of its functional and structural properties, becoming a highly modified extrahaustorial membrane (EHM) that separates the haustorium from the host cytoplasm (Koh *et al*., 2005; O'Connell and Panstruga, 2006). The EHM is tethered to the neck of the haustorium by one or two neck bands (Gil and Gay, 1977), and the sealed compartment thereby created between the EHM and haustorium is called the extrahaustorial matrix (EHMx; Bushnell, 1972). Collectively, the haustorial body and lobes, EHMx and EHM are termed the 'haustorial complex'.

Successful host colonization depends upon the fungus establishing intimate contact with living host cells, which must involve the avoidance or active suppression of plant immune responses. Plant cells are able to sense pathogen attack through a variety of mechanisms, including the perception of pathogen-derived moieties termed PAMPs (pathogen-associated molecular patterns; Schwessinger and Zipfel, 2008), detection of cell wall damage through the release of wall polymer fragments (e.g. oligogalacturonides; Aziz *et al*., 2007) or perturbation of cell wall synthesis, and mechanosensing (reviewed in Hématy *et al.*, 2009). Perception of such pathogen signals may trigger cell wall-associated defence responses that terminate pathogen entry at an early stage, notably through the localized deposition of callose (β-1,3-glucan)-containing papillae (Aist, 1976), coupled with the extracellular generation of reactive oxygen species (Thordal-Christensen *et al.*, 1997; Zeyen *et al.*, 2002) and focal secretion of antimicrobial compounds such as glucosinolates (Bednarek *et al.*, 2009). If these pre-invasion defences are overcome by successful pathogens, the detection of secreted pathogen virulence factors (effectors) inside the plant cytoplasm by cognate resistance proteins (R-proteins) can potentiate a second layer of plant immunity, usually culminating in programmed plant cell death (the hypersensitive response) and a variety of downstream responses mediated by salicylic acid signalling (Micali *et al.*, 2008). Even in susceptible host plants, fungal haustoria may ultimately become sealed-off by a layer of wall-like material secreted by host cells, called a haustorial encasement (Meyer *et al.*, 2009; Wang *et al.*, 2009).

The haustorial interface, comprising the fungal plasma membrane, fungal cell wall, EHMx and EHM, is the principal site for plant–fungal recognition, uptake of nutrients into the pathogen and delivery of secreted effector proteins into host cells (Voegele and Mendgen, 2003; Catanzariti *et al.*, 2006). However, despite its pivotal role, remarkably little is known about the biogenesis of this interfacial zone. Koh *et al.* (2005) proposed two models for formation of the highly specialized EHM, namely invagination coupled with differentiation of the plant plasma membrane, or *de novo* synthesis by targeted vesicle trafficking. Support for the latter hypothesis comes from recent studies on RPW8.2, an Arabidopsis R-protein conferring resistance to powdery mildew fungi (Wang *et al.*, 2009). RPW8.2 is located in the EHM of *G. cichoracearum* but not the host plasma membrane, and is the only EHM-specific component so far identified, apart from an uncharacterized glycoprotein found in the EHM of *Erysiphe pisi* (Roberts *et al.*, 1993). RPW8.2 was also detected in vesicles in the plant cytoplasm adjoining haustoria, suggesting the existence of a dedicated secretory pathway targeting plant proteins to the EHM (Wang *et al.*, 2009). While there is evidence for trafficking of plant components at the haustorial interface, the movement of materials to and from the fungal side of the EHM has not been documented. The composition of the EHM and EHMx and the possible contribution made by the fungus to these structures remains elusive, partly due to technical difficulties associated with isolating haustorial complexes for analysis.

In the present study, we established a method for isolating the haustoria of *G. orontii* from infected Arabidopsis leaf tissue by isopycnic centrifugation. To gain insight into the mechanisms of interface biogenesis and haustorial internalization by host cells, we undertook a comprehensive analysis of the structure and composition of these isolated cells, as well as haustoria *in situ*. We show that molecular specialization of the EHM around *G. orontii* haustoria involves exclusion of arabinogalactan protein epitopes (AGP) normally present in the host plasma membrane and recruitment of the RPW8.2 resistance protein. The exceptional preservation of the plant–fungal interface provided by cryo-fixation TEM (transmission electron microscopy) revealed that host endoplasmic reticulum (ER) is in contact with the EHM over most of its surface, raising the possibility that the ER directly contributes new membrane material to the EHM. We also show that haustorial encasements are densely packed with extracellular plant-derived vesicles, within a matrix of plant cell wall polymers. Finally, we present new evidence for the existence of membrane-bound vesicles in the EHMx and for their possible origin as exosomes derived from fungal multivesicular bodies (MVBs).

Results

Isolation of G. orontii *haustorial complexes from infected* Arabidopsis *leaves*

To facilitate studies on the composition of *G. orontii* haustoria and their associated encasements, we developed a method to isolate them from infected Arabidopsis leaves, based on procedures used previously to purify intracellular hyphae of the bean anthracnose fungus, *Colletotrichum lindemuthianum* (Pain *et al.*, 1994). Powdery mildew colonies proliferate epiphytically (Fig. S1A), with only haustoria entering into epidermal cells (Fig. S1B). To maximize the number of haustoria present in the infected leaves used as starting material, we inoculated *Arabidopsis thaliana* Col-0 plants expressing the bacterial transgene *NahG*. These plants are hypersusceptible to infection by a variety of pathogens due to constitutive catabolism of the defence-signalling molecule salicylic acid (Delaney *et al.*, 1994; Reuber *et al.*, 1998) and support significantly more *G. orontii* haustoria compared with Col-0 wild-type plants. We found that harvesting leaves 4–6 days post inoculation (dpi) maximized the number of haustoria while minimizing contamination from conidia.

After releasing fungal structures from host epidermal cells by homogenization, they were purified by isopycnic centrifugation on Percoll density gradient medium. In preliminary experiments using a discontinuous density gradient, with steps ranging from 0.1 g ml^{-1} to 1.1 g ml^{-1}, the buoyant density of *G. orontii* haustorial complexes was found to vary between 0.71 and 0.86 g ml^{-1}. In all subsequent experiments, samples of tissue homogenate were layered onto Percoll cushions with a density of 0.71 g ml^{-1} so that haustoria entered the Percoll cushion, while plant contaminants such as chloroplasts floated above it and could be removed by aspiration. Haustoria comprised only ~0.5% of the crude leaf homogenate (8.1×10^5 g^{-1} fresh weight of leaf tissue), the main contaminants being plant chloroplasts (1.7×10^8 g^{-1} fresh weight of leaf tissue), fungal conidia (6.1×10^5 g^{-1} fresh weight of leaf tissue) and some broken fragments of epicuticular hyphae and plant cell wall debris (not quantified). After two rounds of isopycnic centrifugation, the haustoria were enriched to ~6% purity, the preparation containing on average 2.7×10^4 haustorial complexes, 4.1×10^5 chloroplasts and 1.3×10^4 conidia per gram fresh weight of leaf tissue (Fig. S1C). The method therefore provided a ~12-fold enrichment of the haustoria from the crude leaf homogenate.

Following isolation, 35–50% of haustoria appeared to retain an intact EHM, as judged by light microscopy. Haustoria were on average 16.2 μm (±2.7 μm) long and 10 μm (±2.1 μm) wide ($n = 60$). Haustorial encasements also appeared intact after isolation and remained attached to the haustorial complexes (Fig. S1D). Approximately 45% of the isolated haustoria were not encased, 26% were partially encased and 29% fully encased ($n = 60$).

Ultrastructure of G. orontii *haustoria and the plant–fungal interface*

To analyse the ultrastructure of haustoria as close as possible to the *in vivo* condition, we used ultra-rapid cryofixation by high-pressure freezing (HPF) followed by freeze substitution (Hippe, 1985). Most Arabidopsis leaf epidermal cells were well preserved although a small number (5–10%) were damaged by ice-crystal formation. However, fungal appressoria, epicuticular hyphae and conidiophores were always lost from the leaf surface during cryo-preparation as they adhered to the HPF sample holders. Each haustorium comprised a narrow neck region containing a septum, subtended by a swollen haustorial body from which numerous thin finger-like projections (haustorial lobes) emerged and eventually coiled around the entire haustorial body. Serial sectioning showed that each haustorial body contained a single nucleus (Fig. 1A), which was confirmed by labelling isolated haustoria with a fluorescent DNA probe (Fig. 6C).

The haustorial body and lobes were separated from the plant cytoplasm by a prominent EHM (Fig. 1B–D), which was thicker and more electron-opaque than the plant plasma membrane lining the plant cell wall. The EHM was asymmetric, the outer leaflet of the lipid bilayer, facing the EHMx, being more electron-opaque than the inner leaflet, facing the host cytoplasm (Fig. 1D). Similar membrane asymmetry was also noted in the EHMs of *Blumeria* and *Erysiphe* (Bracker, 1968; Gil and Gay, 1977). Around young haustoria the EHM was relatively smooth and less electron-lucent, but around mature haustoria it was highly convoluted, with numerous branched evaginations protruding out of the plant cytoplasm towards the haustorium (Figs 1D and S2). The highest complexity of EHM folding occurred near the haustorial neck (Fig. 1E).

Swollen tubules and cisternae of plant ER were consistently located in very close proximity to the EHM, following the contours of the haustorium (Figs 1C and D and S3A and B). When present, ribosomes were asymmetrically distributed on the ER surrounding haustoria, being confined to the ER surface facing the plant vacuole, whereas the surface facing the EHM was devoid of ribosomes (Fig. S3A and B).

In addition to mitochondria, MVBs were among the most abundant organelles in the haustorial cytoplasm. MVBs ranged in size from 200 to 700 nm in diameter, and the contents of their intralumenal vesicles varied in electron opacity (Fig. 2A and B). Single vesicles (approximately 50 nm in diameter) were present in the paramural space between the plasma membrane and cell wall (Fig. 2C). Sites of putative fusion of MVBs with the fungal plasma membrane (Fig. 2D) were observed in two independent haustoria. The EHMx surrounding the haustorial body and lobes was composed of finely granular material with moderate electron opacity. Clusters of membrane-bound vesicles (85–350 nm in diameter) were frequently present within the EHMx close to haustorial lobes (Fig. 2E–G).

Deposits of plant cell wall material were visible at all fungal penetration sites, forming a collar surrounding the haustorial neck region (Figs 1A and 3A). In leaf tissue sampled at 4 dpi, some haustoria were partially encased by a cup-shaped extension of the collar (Fig. 3B), while others were completely surrounded by encasement material (Fig. 3C). These haustorial encasements were always deposited outside the EHM and EHMx (Fig. 3B and C), being bounded by host plasma membrane on both their inner and outer faces and separated from the EHM by a thin layer of host cytoplasm (Fig. 3F and G). A similar topology was described for encasements deposited around haustoria of *Blumeria graminis* f.sp. *hordei* (Bracker, 1968) and the haustoria of rust fungi (Littlefield and Bracker, 1972; Skalamera *et al.*, 1997). All partially and fully encased haustoria that we examined appeared to be mature, as judged from the extent of development of their haustorial lobes, and the encasements varied in thickness from 2.6 to 4 μm (based on measurements from 60 haustoria). Haustorial encasements were highly heterogeneous in structure, comprising numerous membranous inclusions embedded in an electron-lucent matrix. Some inclusions were recognizable as membrane-bound vesicles or MVB-like structures with electron-opaque contents, while others resembled broken or crushed vesicles (Fig. 3D and E). The bounding membrane of these structures displayed the characteristic ‘tramline’ structure of lipid bilayers viewed with TEM

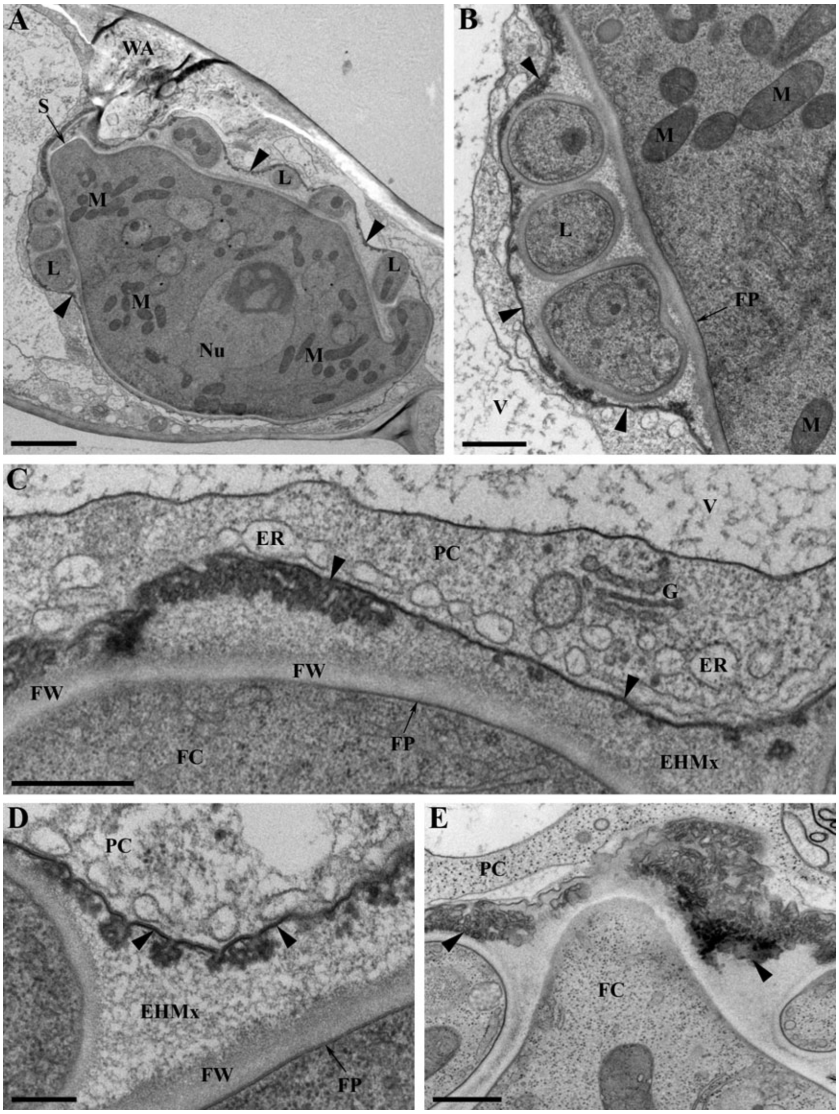

Fig. 1. Ultrastructure of *G. orontii* haustoria infecting *A. thaliana NahG* epidermal cells.
A. Overview of a young haustorium. The haustorial body is separated from the neck by a septum (S) and is surrounded by haustorial lobes (L). The body and lobes are enveloped by the extrahaustorial membrane (EHM, arrowheads). A single nucleus (Nu) and numerous mitochondria (M) are present in the fungal cytoplasm. A plant wall apposition (WA) is located at the haustorial penetration site.
B. Selective enlargement from (A), showing a thin layer of plant cytoplasm between the plant vacuole (V) and EHM (arrowheads).
C. The plant cytoplasm (PC) adjacent to a haustorium contains swollen tubules of endoplasmic reticulum (ER) and Golgi stacks (G). The EHM (arrowheads) is separated from the fungal cell wall (FW) by a finely granular extrahaustorial matrix (EHMx).
D. Detail of the highly convoluted EHM, showing branched outfoldings that protrude into the extrahaustorial matrix (EHMx). The outer leaflet of the EHM facing the EHMx (arrowheads) is more electron-opaque than the inner leaflet.
E. Convolution of the EHM (arrowheads) is greatest around the haustorial neck.
FC, fungal cytoplasm; FP, fungal plasma membrane. Bars: (A) 2 μm; (B, C, E) 500 nm; (D) 200 nm.

(Fig. 3E). The plant cytoplasm close to the 'rim' of haustorial encasements typically contained Golgi bodies and putative secretory vesicles (Fig. 3F and G).

Cytochemical analysis of haustorial encasements

To investigate the composition of *G. orontii* haustorial encasements, we used a panel of eight previously characterized antibodies recognizing protein and polysaccharide components of plant cell walls. These were used for immunofluoresence labelling of unfixed isolated haustorial complexes or TEM-immunogold labelling of high-pressure frozen, resin-embedded haustoria *in planta*. The results are summarized in Table 1.

Using immunofluorescence labelling, β-1,3-glucans, xyloglucans (CCRCM1), rhamnogalacturonan (CCRCM2), arabinogalactan proteins (JIM14 and CCRCM7) and a β-linked galactose-containing protein (JIM1) were all detected on the outer surface of haustorial encasements (Fig. 4A–F). However, these epitopes were generally not detected on the inner face of the encasements, closest to the haustorium. Moreover, JIM14 only labelled the rim and neck region of encasements (Fig. 4D), while CCRCM1 and CCRCM7 preferentially labelled the neck region (Fig. 4C and E). Low- and high-ester pectins (JIM5 and JIM7) and extensins (JIM20) were not detected in encasements (results not shown), and none of the plant cell wall probes labelled the EHM, EHMx or any fungal structures.

TEM-immunogold labelling showed that in uninfected epidermal cells arabinogalactan epitopes recognized by antibody CCRCM7 were restricted to the plant plasma membrane–cell wall interface (Fig. 5A). However, in cells harbouring haustoria, label was distributed throughout the

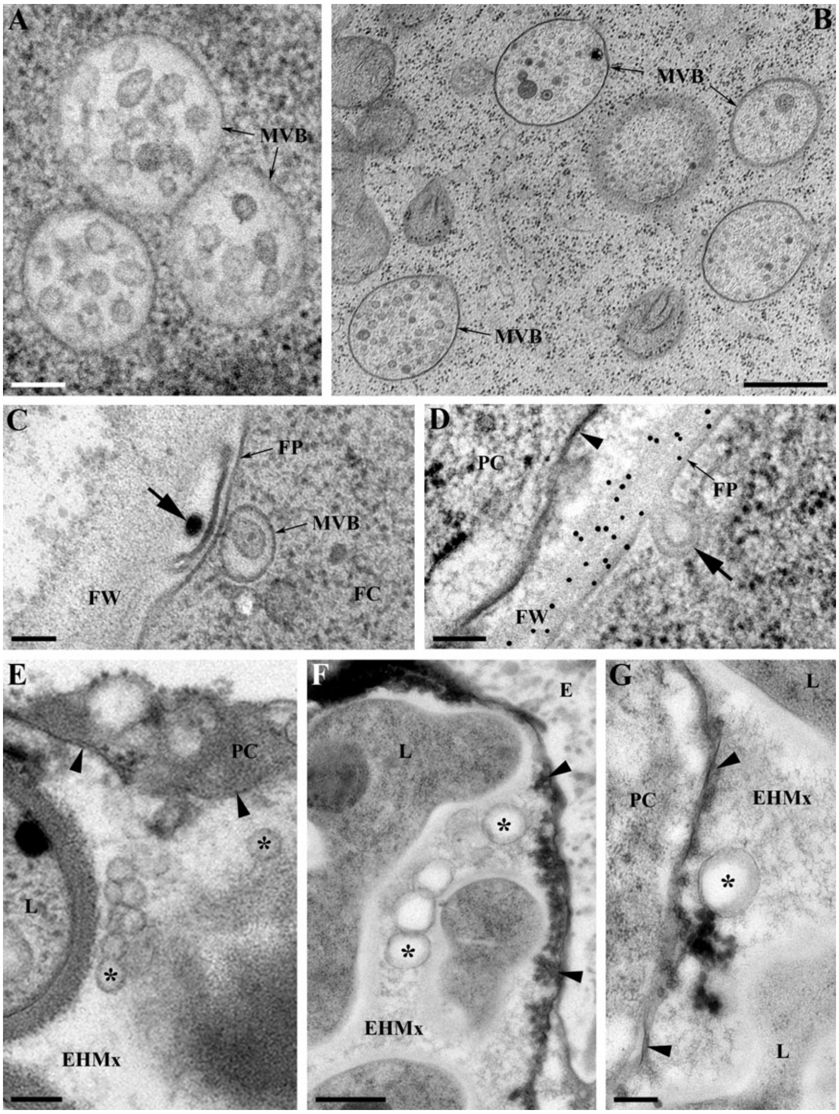

Fig. 2. Vesicular structures in *G. orontii* haustoria.
A and B. Multivesicular bodies (MVB) are among the most prominent organelles in the haustorial body. The content of their intralumenal vesicles varies in electron opacity.
C. An MVB closely associated with the fungal plasma membrane (FP). An electron-opaque vesicle (arrow) is present in the paramural space between the fungal plasma membrane and cell wall (FW).
D. Site of putative vesicle budding/fusion (arrow) at the fungal plasma membrane (FP). The fungal wall (FW) is labelled by anti-β-1,3-glucan antibodies.
E–G. Membrane-bound vesicles (asterisks) are present in the extrahaustorial matrix (EHMx) between haustorial lobes (L) and the extrahaustorial membrane (arrowheads). In (F), notice that the extrahaustorial membrane makes close contact with the haustorial encasement (E).
FC, fungal cytoplasm; PC, plant cytoplasm; arrowheads, extrahaustorial membrane. Bars: (A–E) 100 nm; (F) 500 nm; (G) 200 nm.

plant cell wall surrounding the penetration site, and in papillae and haustorial encasements (Fig. 5B and D). Unlike the normal plant plasma membrane, the EHM was not labelled by CCRCM7 (Fig. 5C). Papillae, collars and haustorial encasements also contained the arabinogalactan protein and β-linked galactose epitopes recognized by JIM14 and JIM1, respectively (Fig. S4C–F), as well as β-1,3-glucans (Fig. 5E and F) and xyloglucan (Fig. S4A and B). All these plant cell wall components appeared evenly distributed through the encasement, with no evidence for layering.

Cytochemical analysis of the haustorial cell wall

To study the composition of the haustorial cell wall we used antibodies recognizing α-1,3-glucans, β-1,3-glucans and chitosan, while the lectin wheatgerm agglutinin (WGA) was used to label chitin. In addition, we used monoclonal antibody UB7, which recognizes a 62 kDa cell wall glycoprotein conserved in many ascomycete fungi that was first identified in *E. pisi* haustoria (Mackie *et al*., 1991; Mitchell *et al*., 1997). Table 1 summarizes the results of this analysis. The cell walls of haustorial bodies and lobes contained both β-1,3-glucans (Figs 6A, S3A and S5A) and chitin (Fig. 6B and C). The cell wall was also strongly labelled by antibody UB7 (Figs 6D and S5B). All epitopes appeared uniformly distributed throughout the haustorial cell wall, except that chitin was not detectable in the region of the haustorial neck band (Fig. 6C). The cell walls of isolated haustorial complexes were only accessible for labelling by antibodies and lectins in those cases where the EHM was absent or damaged. However, chitosan and α-1,3-glucans could not be detected on any isolated haustorial

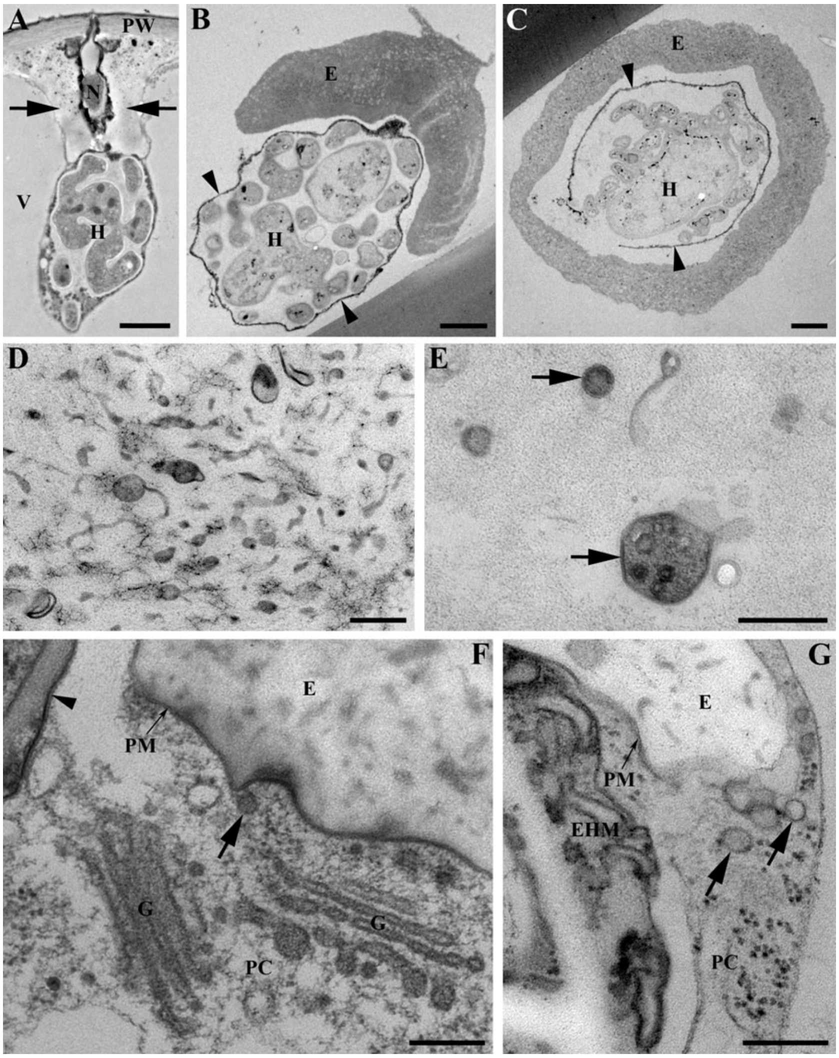

Fig. 3. Development and ultrastructure of callose encasements around *G. orontii* haustoria (A, F, G *in planta*, B–E after isolation).
A. Young haustorium (H) with a callose collar (arrows) surrounding the haustorial neck (N).
B. Partially encased mature haustorium with numerous haustorial lobes. The electron-opaque extrahaustorial membrane (arrowheads) surrounds the haustorium but not the encasement (E).
C. Fully encased mature haustorium showing the encasement is external to the extrahaustorial membrane (arrowheads).
D. Part of an encasement containing numerous electron-opaque membranous inclusions embedded in an electron-lucent matrix.
E. Vesicles and multivesicular bodies embedded in the finely granular matrix of an encasement, showing the bilayer structure of their bounding membranes (arrows).
F and G. The encasement (E) is enveloped by the plant plasma membrane (PM) and putative secretory/endocytic vesicles (arrows). Golgi stacks (G) are present in the plant cytoplasm (PC) adjoining the encasement. Note the plant plasma membrane (PM) and extrahaustorial membrane (EHM) differ in ultrastructure.
PW, plant cell wall; V, plant vacuole. Bars: (A–C) 2 μm; (D–G) 200 nm.

complexes using immunofluorescence microscopy (data not shown). None of the fungal cell wall probes labelled the EHM or EHMx.

Live-cell imaging of plant resistance protein RPW8.2 at the haustorial interface

Recently the plant resistance protein RPW8.2 was shown to be specifically recruited to the EHM of the powdery mildew fungus *G. cichoracearum* (Wang *et al*., 2009). To test whether RPW8.2 was also mobilized to the EHM of *G. orontii*, we infected Arabidopsis Col-0 plants expressing the transgene RPW8.2–YFP under the control of its native promoter (1 kb of 5′ regulatory sequences) (Wang *et al*., 2009) and examined haustoria *in situ* by confocal microscopy at 2 and 4 dpi. RPW8.2–YFP fluorescence was detected almost exclusively at the EHM of mature *G. orontii* haustoria (Fig. 7A, B, E and F). Of 31 haustoria examined, all mature haustoria (20) showed RPW8.2–YFP labelling at the EHM and 90% of them were partially or fully encased, consistent with previous observations in *G. cichoracearum* that suggest RPW8.2 promotes haustorium encasement (Wang *et al*., 2009). In contrast, the EHMs of young, unencased haustoria (11) lacked YFP fluorescence (Fig. 7C and D). RPW8.2–YFP signal was also detected inside haustorial encasements (Fig. 7A, B, E and F) and in papillae (not shown).

Discussion

The EHM and EHMx enveloping powdery mildew haustoria collectively comprise the interface for molecular exchange with host cells, including nutrient uptake, effector delivery, plant–fungal recognition and deployment of plant defence molecules. Despite the importance of the *G. orontii*–Arabidopsis interaction as a model for studying

Table 1. Summary of antibodies and lectins used in this study and the localization of components recognized within *G. orontii* haustorial complexes.

Probe	Component recognized	Localization
JIM1 (m)	β-linked galactose-containing plasma membrane protein[a]	Encasements, papillae
JIM5 (m)	Unesterified pectins[b]	No signal
JIM7 (m)	Methyl-esterified pectins[b]	No signal
JIM14 (m)	Arabinogalactan protein[b]	Encasements, papillae
JIM20 (m)	Extensin[c]	No signal
UB7 (m)	Fungal cell wall glycoprotein[d]	Haustorial cell wall
UBIM22 (m)	Rat bone cells[d]	No signal
CCRCM1 (m)	Fucosylated xyloglucan, rhamnogalacturonan I[e]	Encasements, papillae, plant cell wall
CCRCM2 (m)	Rhamnogalacturonan I[e]	Encasements, papillae
CCRCM7 (m)	Rhamnogalacturonan I, arabinogalactan protein[e]	Encasements, papillae, plant plasma membrane
MOPC-104E (m)	α-1,3-glucan[f]	No signal
β-1,3-glucan (m)	Linear β-1,3-glucan, callose[g]	Haustorial cell wall, encasements, papillae
Chitosan (p)	Deacetylated chitin[h]	No signal
Wheatgerm agglutinin	β-1,4-*N*-acetylglucosamine, chitin oligosaccharides[i]	Haustorial cell wall

Results are based on at least three replicate experiments per probe. Publications reporting the binding specificities of monoclonal (m) and polyclonal (p) antibodies are as follows: **a.** Knox and Roberts, 1989; **b.** Knox *et al.*, 1990; **c.** Smallwood *et al.*, 1994; **d.** Mackie *et al.*, 1991; **e.** Puhlmann *et al.*, 1994; **f.** Rappleye *et al.*, 2007; **g.** Meikle *et al.*, 1991; **h.** El Gueddari *et al.*, 2002; **i.** Ramonell *et al.*, 2005.

biotrophic parasitism, no information is available on the haustorial interface. Here, we used haustoria isolated from infected plants, cryo-fixation TEM and immunocytochemistry to dissect the structure and composition of this interface. We show that specialization of the EHM involves exclusion of a plasma membrane AGP and the developmentally regulated recruitment of the RPW8.2 protein. We also demonstrate the entrapment of extracellular plant-derived vesicles and MVBs within haustorial encasements and present the first evidence for a putative exosome-mediated secretory pathway in haustoria.

Haustorial complex isolation: utility and applications

To separate haustoria away from other fungal structures and contaminating plant components we used isopycnic centrifugation on an osmotically inert colloidal silica density gradient medium, similar to procedures previously used to isolate haustoria and intracellular hyphae of other fungi (Gil and Gay, 1977; Mackie *et al.*, 1991; Pain *et al.*, 1994). We obtained samples of intact *G. orontii* haustorial complexes with a purity of 6% and yield of approximately 3×10^4 haustoria per gram fresh weight of leaf tissue. Such partially purified preparations can be used for cell biology studies, as well as for transcriptome and proteome analyses. Although plant chloroplasts were the most abundant contaminating particles, chloroplast RNA is not polyadenylated (Wheeler and Hartley, 1975) and therefore would not be selected during oligo-dT purification of mRNA prior to cDNA library construction. Preliminary EST sequencing indicated that total RNA from preparations obtained in this study comprises 85% fungal sequences, assembling into approximately 30 000 contigs that correspond to more than 15 000 transcripts (E. Ver Loren van Themaat and R. Panstruga, unpublished). Proteomic analysis of haustoria would require further enrichment by affinity purification, e.g. using fungal wall or EHM-specific markers for fluorescence-activated cell sorting (FACS), as recently reported for *Colletotrichum higginsianum* (Takahara *et al.*, 2009). The plant resistance protein RPW8.2 would be a good candidate to mark haustoria for FACS-based enrichment, since it accumulates at the EHM of *G. orontii* (this study) and *G. cichoracearum* (Wang *et al.*, 2009).

Cell wall PAMPs are exposed at the surface of G. orontii *haustoria*

As haustoria develop in intimate contact with living host cells, molecules derived from fungal wall polymers such as chitin and glucans are likely to be exposed or released and act as PAMPs, potentially triggering plant immune responses. For example, chitin and chitin oligosaccharides are recognized by plant immune receptors and can trigger defence responses in both monocot and dicot plants (Miya *et al.*, 2007; Wan *et al.*, 2008). As counter-defence strategies, some plant pathogens secrete chitin-binding proteins to mask chitin from attack by plant chitinase enzymes (van Esse *et al.*, 2007) or sequester released chitin fragments to evade host recognition (de Jonge *et al.*, 2010). Alternatively, surface-exposed chitin can be converted to chitosan through the action of chitin deacetylase enzymes (El Gueddari *et al.*, 2002).

Chitin and β-1,3-glucans were previously detected in the walls of powdery mildew haustoria in cross-sectioned cells (Hajlaoui *et al.*, 1991; Celio *et al.*, 2004; Ramonell *et al.*, 2005) and in fixed/permeabilized cells (Yun *et al.*, 2003). Through the analysis of isolated haustoria, we were able to show that both these polysaccharides are exposed on the fungal cell surface, although epitopes were only accessible

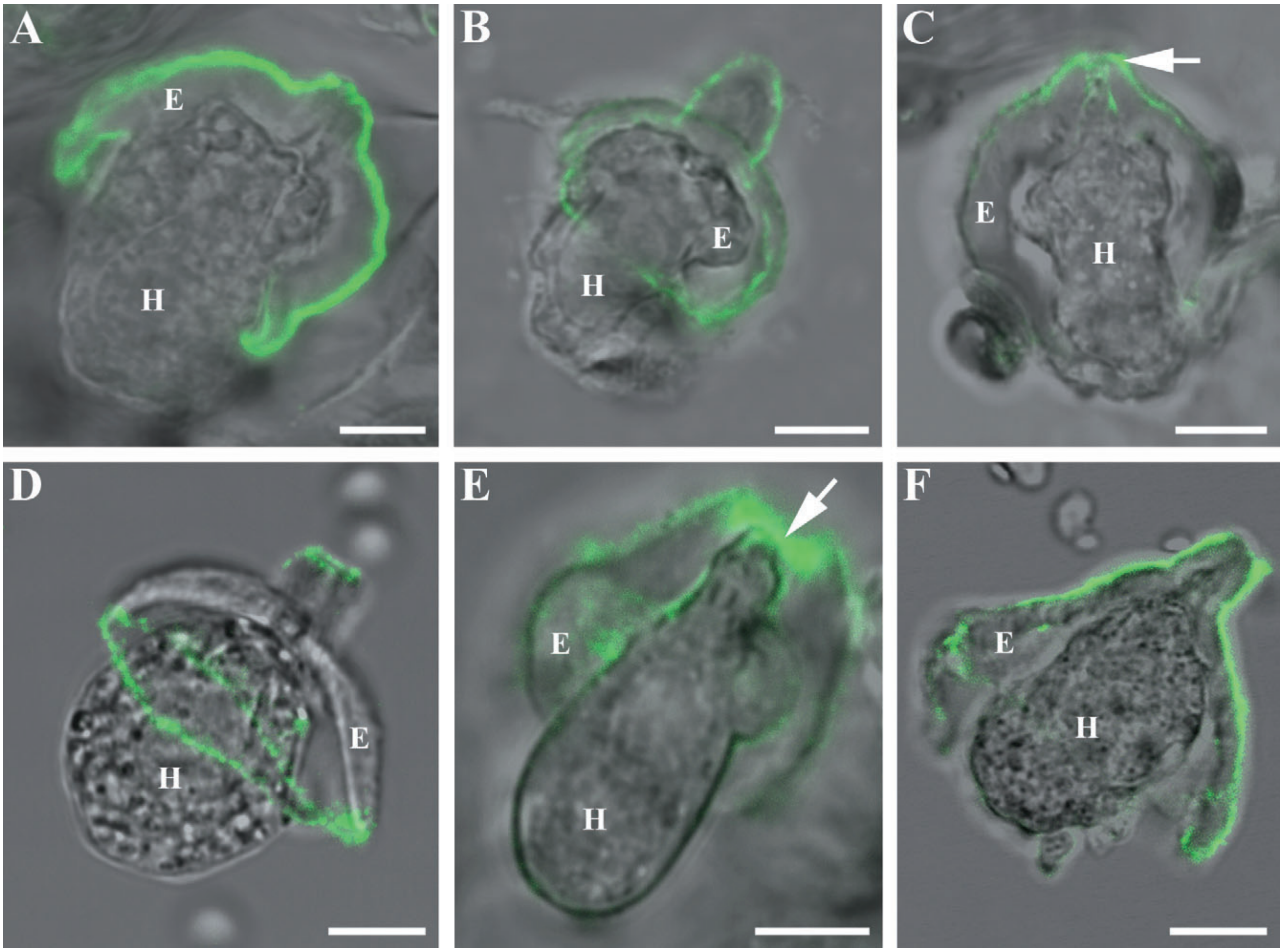

Fig. 4. Localization of plant cell wall components in isolated haustorial complexes of *G. orontii*. Confocal laser scanning micrographs (presented as overlays of bright-field and fluorescence channels) show immunofluorescence labelling with antibodies recognizing:
A. β-1,3-Glucan. Notice absence of labelling of the fungal cell wall, due to the presence of an intact EHM which blocks antibody access to the haustorium surface.
B. Rhamnogalacturonan 1 (CCRCM2).
C. Xyloglucan (CCRCM1).
D and E. Arabinogalactan proteins (JIM14, D; CCRCM7, E).
F. β-Linked galactose (JIM1).
All epitopes were detected on the outer face of the encasement (E), but not on the inner face adjoining the haustorium (H). JIM14 epitopes were only detected at the rim and neck regions of encasements, while CCRCM1 and CCRCM7 preferentially labelled the neck (arrows). None of the probes labelled the extrahaustorial membrane. Bars: 5 μm.

for labelling in those cases where the EHM was absent or damaged. We also detected a 62 kDa glycoprotein recognized by antibody UB7 on the surface of haustoria. This antigen occurs in the cell wall and plasma membrane of *E. pisi* (Mackie *et al*., 1991) and appears to be a conserved component of ascomycete cell walls (Mitchell *et al*., 1997), but its activity as a fungal PAMP has not been determined. We found no evidence for the conversion of chitin to chitosan, as was reported for post-penetration infection structures of the rust fungi *Puccinia graminis* and *Uromyces fabae* and the anthracnose fungus *Colletotrichum graminicola* (El Gueddari *et al*., 2002). Thus, antibodies against chitosan failed to label the surface of *G. orontii* haustoria, whereas chitin was consistently detected at the haustorial cell wall using WGA.

Although chitin deacetylase-like sequences were detected in the draft genome and transcriptome of *G. orontii* (E. Ver Loren van Themaat and R. Panstruga, unpublished), our data suggest these enzymes do not play a significant role in chitin modification in haustorial cell walls. Moreover, we found no evidence that β-1,3-glucans are over-layered by α-1,3-glucans on the surface of haustorial walls. In contrast, surface-localized α-1,3-glucans were shown to mask underlying β-1,3-glucans in the infection hyphae of the rice blast fungus *Magnaporthe grisea* and yeast cells of the human pathogen

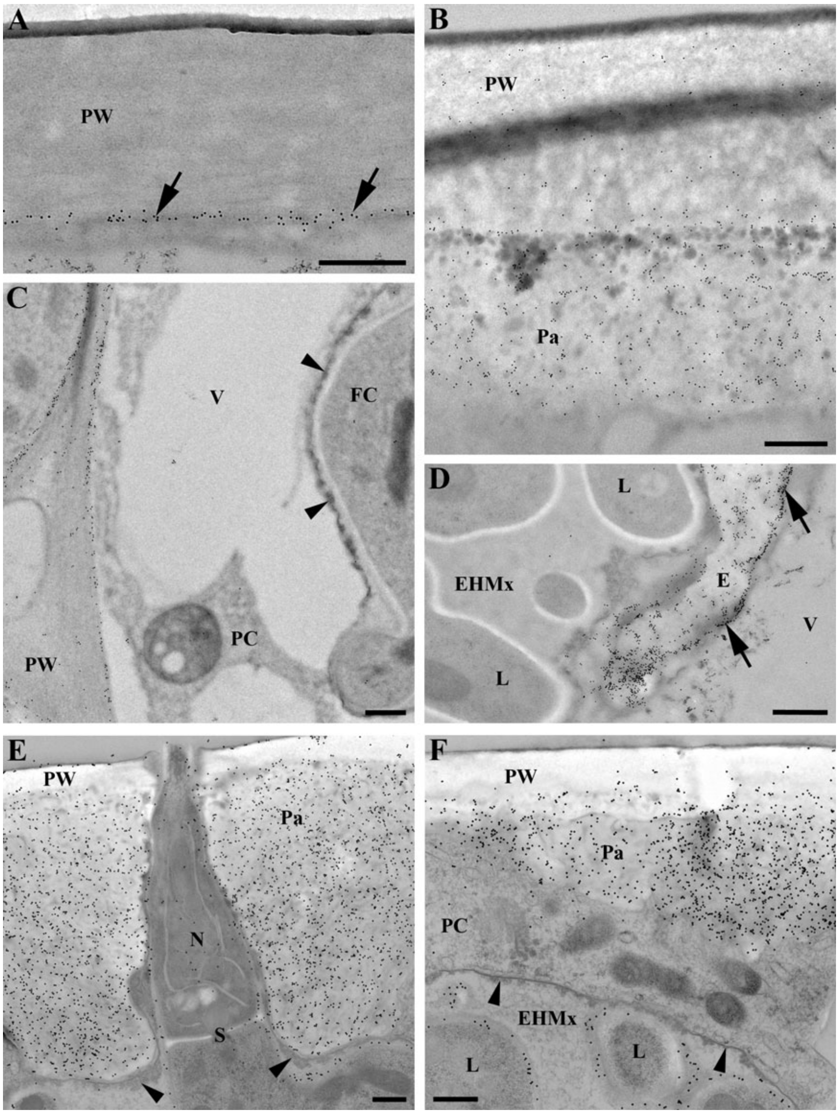

Fig. 5. Immunogold labelling of arabinogalactan protein (AGP) (A–D) and β-1,3-glucan epitopes (E and F) in *G. orontii* haustoria *in planta*.
A. In uninfected *A. thaliana NahG* epidermal cells, antibody CCRCM7 labels AGPs in the plant plasma membrane (arrows).
B. In cells penetrated by *G. orontii* haustoria, CCRCM7 epitopes are detected in both the plant cell wall (PW) and papilla (Pa) around the fungal penetration site.
C. While CCRCM7 labels the plant plasma membrane adjoining anticlinal cell walls, the extrahaustorial membrane (arrowheads) is not labelled.
D. The haustorial encasement (E) is strongly labelled by CCRCM7, especially near the plant plasma membrane (arrows).
E and F. β-1,3-Glucan antibodies strongly label papillae (Pa) around the haustorial neck (N) and cell walls of haustorial lobes (L), but the extrahaustorial membrane (arrowheads) and extrahaustorial matrix (EHMx) are not labelled.
FC, fungal cytoplasm; PC, plant cytoplasm; S, septum, V, plant vacuole. Bars: 500 nm.

Histoplasma capsulatum (Rappleye *et al.*, 2007; Fujikawa *et al.*, 2009). Concealment of β-1,3-glucan by α-1,3-glucan contributes to the virulence of *Histoplasma* by allowing the pathogen to evade detection by the human immune system via the β-glucan receptor dectin-1 (Rappleye *et al.*, 2007). Thus, we found no evidence for either the chemical conversion or masking of several known cell wall PAMPs in the haustoria of *G. orontii*, suggesting that the pathogen uses other mechanisms to evade or suppress PAMP-triggered immune responses.

Haustorial encasements contain membrane-bound vesicles and plant wall polymers

Encasement is a frequent fate for *G. orontii* haustoria in Arabidopsis; we found that 55% of haustoria isolated from infected plants were either partially or fully encased. Haustorial encasements develop as extensions from papillae, starting as a collar around the haustorial neck that expands to partially or completely enclose the haustorial body (Bushnell, 1972; Meyer *et al.*, 2009). Papillae form rapidly in response to wounding or penetrating pathogens through the localized deposition of materials between the plant plasma membrane and cell wall (Aist, 1976). They are agglomerations of callose (β-1,3-glucan) and many other plant-derived components, which may include cellulose, hemicelluloses, pectins, HRGPs, AGPs, silicon and phenolic materials such as lignin, depending on the plant–pathogen interaction (Hajlaoui *et al.*, 1991; Stark-Urnau and Mendgen, 1995; Zeyen *et al.*, 2002; Celio *et al.*, 2004). Papillae are also sites for the focal accumulation of antimicrobial peptides, toxic secondary metabolites and reactive oxygen species (Thordal-Christensen *et al.*, 1997; Zeyen *et al.*, 2002).

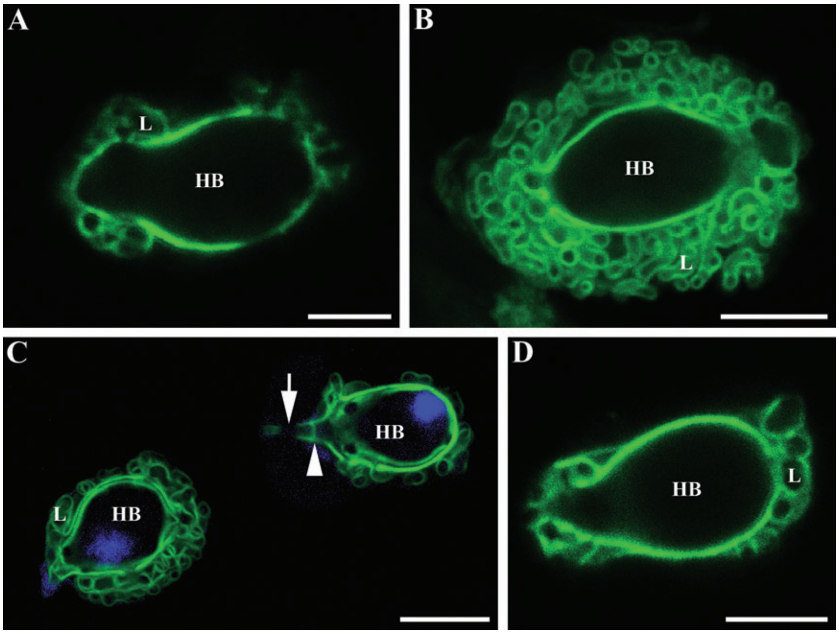

Fig. 6. Localization of fungal cell wall polymers in isolated haustorial complexes of *G. orontii*. Confocal laser scanning micrographs show immunofluorescence labelling with antibodies recognizing:
A. β-1,3-Glucan. Notice abundant labelling of fungal cell wall in contrast to absence of labelling when haustoria carry an intact EHM (see Fig. 4A).
B and C. Wheatgerm agglutinin recognizing chitin. In (C), wheatgerm agglutinin labels the septum in the haustorial neck (arrowhead) but not a region corresponding to the haustorial neck band (arrow). A single nucleus in the haustorial body is stained by DAPI.
D. Antibody UB7 recognizing a fungal N-linked cell wall glycoprotein.
Cell walls of the haustorial body (HB) and lobes (L) are labelled by all three probes.
Bars: 5 μm.

Although papillae have been extensively studied, much less is known about the composition and function of haustorial encasements. We reported previously that encasements are still formed in Arabidopsis *pmr4* mutants deficient in the GSL5 callose synthase, indicating that encasements must contain other cell wall polymers in addition to callose (Meyer *et al*., 2009). Consistent with this, we show here that *G. orontii* encasements contain many components of plant plasma membranes and primary cell walls, including AGPs and glycosylated plasma membrane proteins distinct from AGPs (see Knox and Roberts, 1989), xyloglucans and rhamnogalacturonans. These components were uniformly distributed throughout the thickness of *G. orontii* encasements, in contrast to papillae and haustorial encasements formed in other plant–fungal interactions which often display a multilayered structure (Zeyen and Bushnell, 1979; Zeyen *et al*., 2002), with particular plant glycoproteins and polysaccharides deposited sequentially into different layers (Hajlaoui *et al*., 1991; Stark-Urnau and Mendgen, 1995; Celio *et al*., 2004).

We detected no differences in composition between papillae, collars and encasements with the cytochemical probes used here. Likewise, all of these extracellular deposits had a similar ultrastructure, comprising abundant membrane inclusions, intact vesicles and MVBs embedded within an amorphous matrix (Fig. 3D and E). Their strikingly similar structure and composition suggests that a shared mechanism of targeted host secretion produces both papillae and haustorial encasements. The presence of embedded membranous material corroborates previous findings that both papillae and haustorial encasements label with the lipophilic dye FM4-64 (Meyer *et al*., 2009) and supports the hypothesis that encasement involves the extracellular transport and entrapment of secretory vesicles and their cargo at the growing rim of the cup-shaped encasement. We previously proposed that encasement biogenesis involves exosome-mediated secretion (Meyer *et al*., 2009), and earlier ultrastructural studies also implicated plant MVBs and exosomes in papilla formation (An *et al*., 2006a,b). In the present study, we found Golgi bodies and secretory vesicles were frequently located in the plant cytoplasm adjacent to the encasement rim (Fig. 3F and G), raising the possibility that conventional exocytosis also contributes to encasement formation.

Haustorial encasements could function in plant defence in several ways. By acting as a permeability barrier, they may restrict the uptake of nutrients and water into the parasite and delivery of fungal effectors into host cells. Second, encasements may serve as a matrix to concentrate and compartmentalize plant-derived antimicrobial compounds at the plant–fungal interface, thereby poisoning the haustorium while protecting the host cell from damage by cytotoxic molecules (Wang *et al*., 2009). Several plant plasma membrane proteins potentially involved in the secretion of antimicrobial compounds are selectively incorporated into encasements, including the PEN1 syntaxin, its interacting partner SNAP33 and the PEN3 ABC transporter (Meyer *et al*., 2009). While PEN1 mediates the polarized secretion of unknown cargo that is required for penetration resistance to powdery mildews, PEN3 is implicated in the efflux of glucosinolates, a class of secondary metabolites with broad-spectrum antifungal

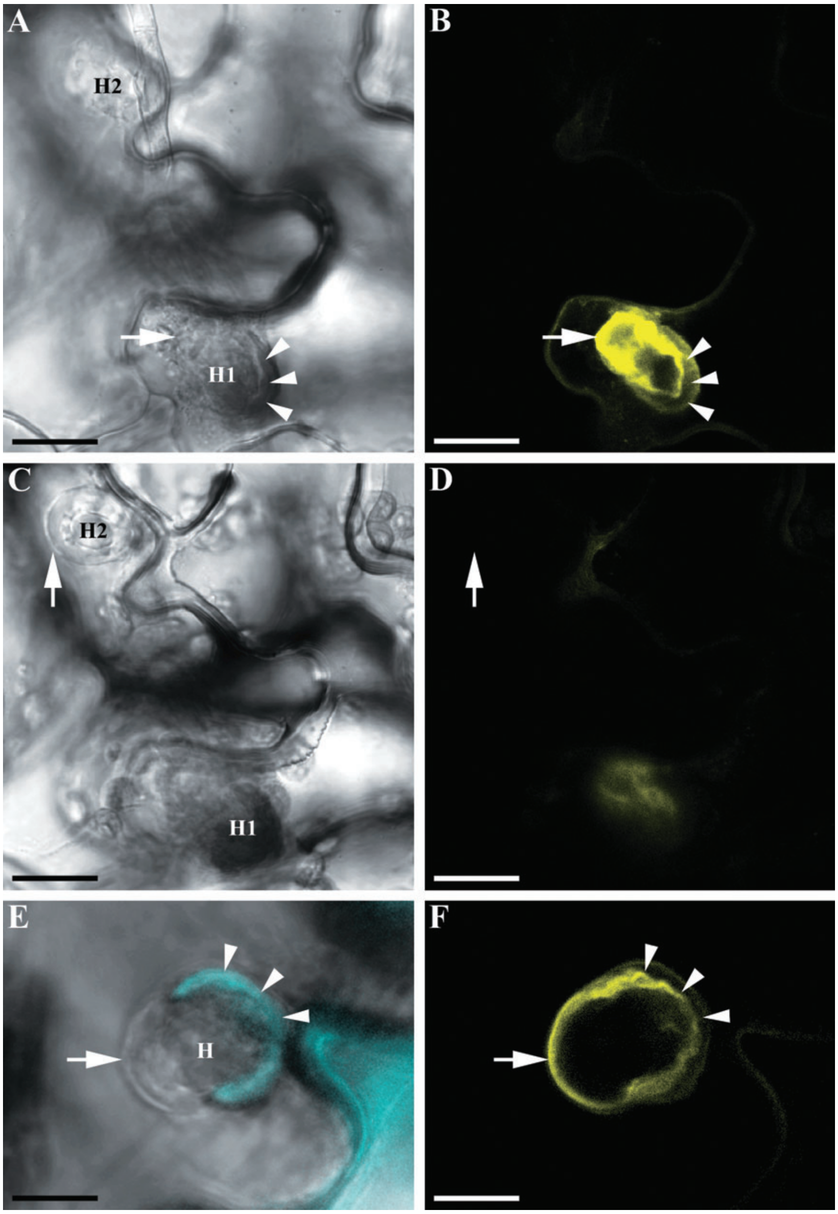

Fig. 7. Arabidopsis resistance protein RPW8.2 localizes to the *G. orontii* EHM in a developmentally dependent manner. Arabidopsis Col-0 plants carrying the transgene RPW8.2–YFP under control of native 5′ regulatory sequences were infected with *G. orontii* and haustoria were inspected by confocal microscopy at 2 dpi (A–D) and 4 dpi (E and F). Bars: 5 μm.
A and B. The RPW8.2–YFP signal is visible at the EHM (arrow) of the primary, more mature haustorium (H1). Some YFP label is also evident in a cup-shaped encasement (arrowheads).
C and D. Different focal plane of the cells shown in (A) and (B). The younger, secondary haustorium (H2) is not encased and lacks YFP label at the EHM (arrow).
E and F. Four days after inoculation, a mature haustorium is partially encased in callose (Aniline blue staining, arrowheads) and shows strong RPW8.2–YFP labelling at its EHM (arrow).

activity (Bednarek *et al*., 2009). Our haustorial isolation method should facilitate future analysis of the secondary metabolite composition of encasements, which may provide further insights into mechanisms of post-penetration resistance to powdery mildews.

Specialization of the G. orontii *EHM*

Earlier studies on the EHM of several species of rust and powdery mildew fungi have revealed its unique structure and composition. Thus, freeze-fracture electron microscopy indicated that transmembrane proteins are absent from the EHM, which also lacks the H^+-ATPase activity associated with normal plant plasma membranes (reviewed in O'Connell and Panstruga, 2006). More recently, GFP tagging was used to show that eight plasma membrane marker proteins are all absent from the EHM of *Erysiphe cichoracearum*, leading to the suggestion that the EHM is either a highly modified plant plasma membrane or a *de novo* generated structure (Koh *et al*., 2005). In the present study, we show that plasma membrane-resident proteins such as AGPs and non-AGP glycoproteins are also absent from the *G. orontii* EHM.

The EHM of *G. orontii* is a dynamic interface, whose structure and composition changes with the developmental stage of the haustorium. Thus, the EHM progressed from a relatively smooth, lipid bilayer in young haustoria to a convoluted, thickened structure in mature haustoria, where the membrane convolutions were most highly elaborated near the haustorial neck. Similar evaginations of the EHM have been noted in the haustoria of powdery mildew species infecting both monocot and dicot plants (Bracker, 1968; Gil and Gay, 1977; Hippe, 1985; Celio *et al.*, 2004). These membrane structures significantly expand the surface area available for nutrient and water uptake by the pathogen and translocation of effectors and other virulence factors into host cells. Membrane expansion and convolution during EHM maturation could result from an imbalance between the delivery of new membrane and recycling through endocytosis.

Tubules and cisternae of host ER were located in very close proximity to the *G. orontii* EHM; indeed the bounding membranes of EHM and ER were frequently so close that they could not be resolved. Previous confocal microscope studies have also indicated a close proximity or even transient physical association between these two structures around haustoria of *E. pisi* and *G. cichoracearum* (Leckie *et al.*, 1995; Koh *et al.*, 2005). Thus, in *E. pisi* haustoria, portions of the host ER remained adhering to the EHM following detergent extraction, suggesting that the ER may be tethered to the EHM at multiple locations (Leckie *et al.*, 1995). Conceivably this arrangement could allow the direct transfer of membrane lipids and proteins from the ER to the developing EHM, without the involvement of vesicle trafficking, as proposed by Leckie *et al.* (1995).

In addition to contributions from the ER, the EHM may mature through the targeted delivery of host-derived vesicles containing specific proteins and lipid components (Koh *et al.*, 2005). Thus, we found that the plant resistance protein RPW8.2 was mobilized to the EHM of *G. orontii* haustoria, as reported previously for *G. cichoracearum* (Wang *et al.*, 2009). However, in the case of *G. orontii*, RPW8.2 was only detectable in EHMs of mature and encased haustoria, but not in EHMs of young haustoria. This may indicate that *G. orontii* is able to delay RPW8.2 expression and/or mobilization to the EHM. Alternatively, the signal required for RPW8.2 expression/delivery may only be triggered once the haustorium is mature. We also detected RPW8.2 within *G. orontii* haustorial encasements and papillae, suggesting vesicles containing the resistance protein become entrapped within these structures. However, previous results with *G. cichoracearum* indicate that RPW8.2 expression has no measurable impact on penetration resistance (Wang *et al.*, 2009). The dynamics of RPW8 incorporation into the EHM and encasements and its role in defence against *G. orontii* require further analysis. However, our findings demonstrate that the composition of the EHM evolves during haustorium maturation through the incorporation of plant-derived components.

Fungal MVBs and extrahaustorial vesicles may constitute an exosome-mediated secretion pathway

The existence of MVBs within fungal cells is well documented and the ESCRT (Endosomal Sorting Complex Required for Transport) machinery responsible for their formation and regulation has been well studied in yeast (reviewed in Schorey and Bhatnagar, 2008). Earlier studies have reported the presence of MVBs in epicuticular hyphae, appressoria and haustoria of the barley powdery mildew fungus, *B. graminis* f.sp. *hordei* (Hippe, 1985; Hippe-Sanwald *et al.*, 1992; An *et al.*, 2006a). Here, we found that MVBs containing intralumenal vesicles of varying electron opacity were one of the most abundant organelles in the cytoplasm of *G. orontii* haustoria. Moreover, we observed MVB-like structures apparently fusing with (or budding off from) the fungal plasma membrane, and single vesicles were present within the paramural space between the fungal plasma membrane and cell wall. Paramural vesicles were noted at the tips of *Blumeria* appressoria during host penetration (Zeyen and Bushnell, 1979; Zeyen *et al.*, 2002; An *et al.*, 2006a), but they were not previously described in powdery mildew haustoria. Importantly, we also detected clusters of vesicles within the EHMx that were clearly delimited by a lipid bilayer (Fig. 2E–G). To our knowledge, membrane-bound vesicles have never been reported in the EHMx of any haustorium-forming phytopathogen.

Although MVBs function in protein and membrane recycling and are often destined for degradation within lysosomes/vacuoles, they can also follow a retrograde pathway involving fusion with the plasma membrane, thereby delivering their intralumenal vesicles to the cell exterior, where they are called exosomes (Schorey and Bhatnagar, 2008). In the human fungal pathogen *Cryptococcus neoformans*, exosomes containing the major capsular polysaccharide glucuronoxylomannan and other virulence factors, such as laccase, urease, superoxide dismutase and glucosylceramide, have been isolated from the extracellular milieu (Rodrigues *et al.*, 2008a). Exosomes were also isolated from culture supernatants of other human pathogens such as *H. capsulatum*, *Candida albicans* and *C. parapsilosis* (Albuquerque *et al.*, 2008), and have been proposed as an alternative route for secretion of virulence and pathogenicity factors to the host (Rodrigues *et al.*, 2008b; Casadevall *et al.*, 2009). In *G. orontii*, the EHMx vesicles varied in size from 50 to 300 nm, comparable to the size range reported for exosomes of human fungal pathogens (10–350 nm; Rodrigues *et al.*, 2008b). Debate exists over the porosity of the fungal cell wall and the ability of membranous vesicles to cross this barrier. However, in both basidiomycete and ascomycete pathogenic yeasts, vesicular structures have been observed within cell walls using TEM (Albuquerque *et al.*,

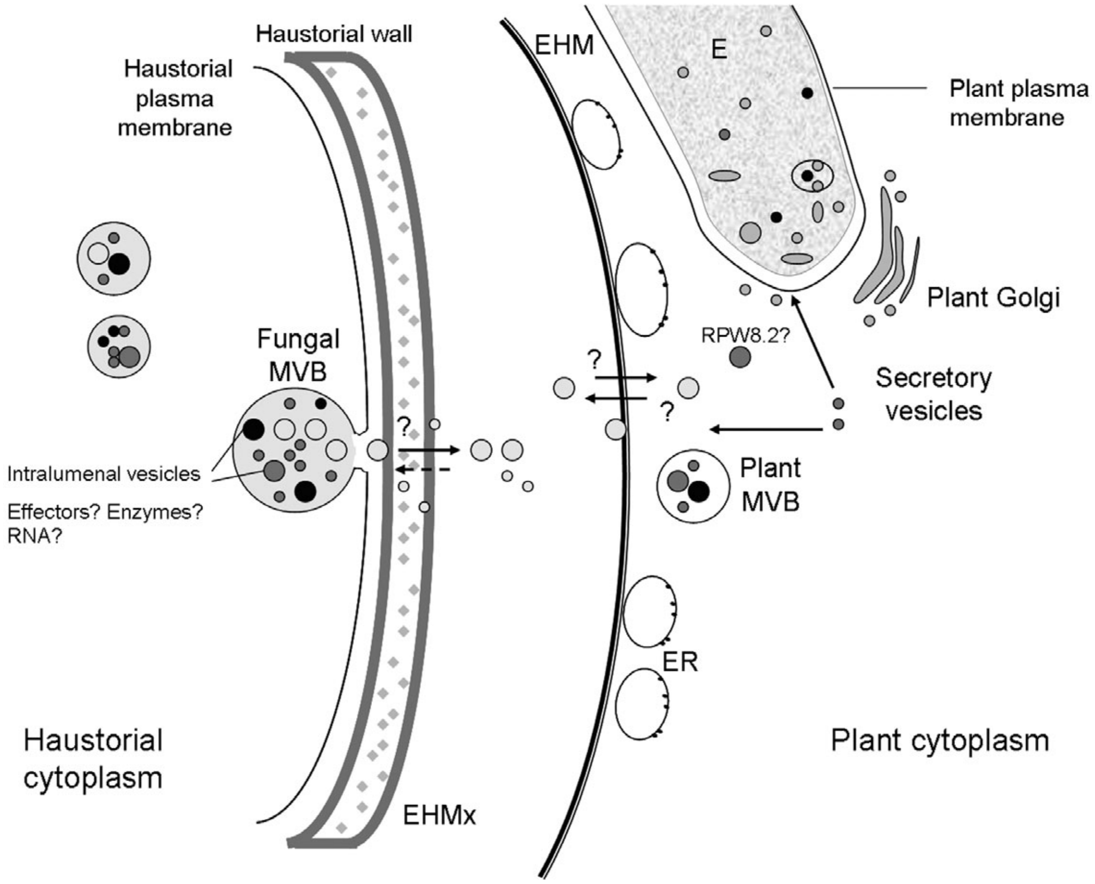

Fig. 8. Schematic diagram depicting putative vesicle dynamics at the plant–fungal interface of *G. orontii*–Arabidopsis. Fusion of fungal MVBs with the plasma membrane releases intralumenal vesicles as fungal exosomes destined for delivery outside the haustorial cell wall, to the EHMx or to further fuse with the EHM. Alternatively, vesicles in the EHMx may originate by budding from the EHM. On the plant side, vesicles originating from Golgi and MVBs fuse with the EHM to deliver their cargo, e.g. resistance protein RPW8.2. Plant secretory vesicles and MVBs are also delivered to the haustorial encasement (E).

2008; Casadevall *et al*., 2009), while in *Saccharomyces cerevisiae*, cell wall pores varying in size from 200 to 400 nm were reported (De Souza Pereira and Geibel, 1999), which would be large enough to allow the passage of vesicles of the size seen in the *G. orontii* EHMx.

On the basis of TEM evidence alone, and in the absence of a specific molecular marker, we cannot determine whether the vesicles observed in the EHMx represent exosomes derived from the fusion of fungal MVBs with the haustorial plasma membrane, or plant vesicles produced by budding-off from the EHM. However, it is tempting to speculate that they represent an intermediate stage in the delivery of fungal virulence factors to the host cell. We thus propose a model in which fungal exosomes would pass from the paramural space, across the fungal cell wall to enter the EHMx. There, the vesicles could rupture and release their contents into the matrix, or alternatively fuse with the EHM, thereby delivering their cargoes directly into the plant cytoplasm (Fig. 8). The possible existence of such an alternate vesicle transport route, distinct from the canonical ER/Golgi-based secretion pathway, for the delivery of pathogen effectors to the plant cytoplasm is especially pertinent in the case of powdery mildew fungi, since the only two known avirulence proteins, AVRk1 and AVRa10, both lack a canonical N-terminal secretion signal (Ridout *et al*., 2006).

Experimental procedures

Plant and fungal material

Arabidopsis thaliana plants were grown as described previously (Göllner *et al*., 2008). Plants used in this study were the wild-type Col-0 accession and transgenic lines expressing either *NahG* (Delaney *et al*., 1994) or the RPW8.2–YFP fusion protein under the native RPW8.2 promoter (Wang *et al*., 2009), both in the Col-0 genetic background. Four- to 5-week-old plants were inoculated with conidia of *G. orontii* isolate MPIZ by brushing rosette leaves against heavily infected plants and incubated as described previously (Göllner *et al*., 2008). For confocal microscopy, the cotyledons of 10-day-old seedlings were inoculated and examined at 2 or 4 dpi.

Isolation of G. orontii *haustorial complexes by isopycnic centrifugation*

The method used for haustorial isolations was modified from that of Pain *et al*. (1994). Prior to harvesting haustoria at 4–6 dpi, infected plants were transferred to dark conditions for 36 h, which was found to facilitate the subsequent removal of chloroplasts by density centrifugation by reducing the size of starch grains, and thus chloroplast buoyant density. For extractions, approximately 20 g of infected leaves were excised, washed in an aqueous solution of 1% (w/v) Tween 20 detergent, to remove epiphytic mycelium and conidia and rinsed extensively with distilled water. Leaves were then blotted dry and homogenized for 30–40 s in 100 ml of ice-cold isolation buffer [0.02M MOPS (3-N-morpholino propane sulfonic acid), 0.2 M sucrose, pH 7.2] in a pre-chilled kitchen blender (Krups mini-chopper GVA2, 400W, Solingen, Germany). All subsequent steps of the procedure were performed at 4°C. The slurry was filtered through a 50 μm nylon mesh and the residue retained on the mesh was re-homogenized in 100 ml of isolation buffer and re-filtered. The filtrates were combined, re-filtered and centrifuged at 1080 *g* for 15 min in a bench centrifuge with a swing-out rotor. Pellets were resuspended in 20 ml of buffer and 5 ml aliquots were layered onto 5 ml of Percoll solution (GE Healthcare Bio-sciences, Uppsala, Sweden) diluted to a density of 0.86 g ml^{-1} in isolation buffer. After centrifugation at 720 *g* for 15 min, the upper layer and interface containing chloroplasts and other debris was discarded and each Percoll layer was diluted to 50 ml with buffer. The haustorial complexes in this fraction were collected by centrifugation at 1080 *g*, the pellets were pooled and re-suspended in 10 ml of buffer and the isopycnic centrifugation step was repeated. The final Percoll fraction was diluted to 50 ml with buffer and centrifuged at 1080 *g*. The pellet was re-suspended in 500 μl of buffer and the number of haustorial complexes, plant organelles and other fungal structures were estimated using a haemocytometer.

Differential interference contrast (DIC) microscopy

For differential interference contrast (DIC) microscopy of *G. orontii* infection structures, Arabidopsis leaves and cotyledons were decolourized in a 3:1 mixture of ethanol and chloroform for 24 h and then cleared in saturated chloral hydrate solution for 24 h. Samples were mounted in saturated chloral hydrate and viewed with a Zeiss Axio Imager Z1 (Carl Zeiss, Jena, Germany). Images were processed using Adobe Photoshop 10.0.

Cytochemical labelling and confocal microscopy

The specificities of the probes used for cytochemical labelling are presented in Table 1. Monoclonal antibodies recognizing α-1,3-glucans and β-1,3-glucans were purchased from Abcam (Cambridge, UK) and Biosupplies Australia Pty. (Parkville, Australia) respectively. Haustorial complexes were applied to multi-well microscope slides coated with poly-L-lysine to promote cell attachment and air-dried overnight. Prior to labelling, slides were rinsed in phosphate-buffered saline (PBS). For antibody labelling, the samples were incubated for 30 min in block buffer [5% v/v normal goat serum (Sigma) and 6% w/v bovine serum albumin (Sigma) in PBS]. Primary antibody, neat or diluted 1:10 or 1:20 in PBS, was then applied to the sample and incubated for 1–2 h in a moist chamber at room temperature. After washing in PBS for 3 × 15 min, samples were incubated with a 1:50 dilution of the appropriate secondary antibody conjugated to fluorescein isothiocyanate (FITC, Sigma) for 1 h and rinsed in PBS for 3 × 15 min. For chitin labelling, samples were incubated with WGA conjugated to FITC (Sigma) diluted to 100 μg ml^{-1} in PBS at room temperature in the dark for 30 min and rinsed with PBS. Nuclei were stained with 2.5 μg ml^{-1} DAPI (4,6-diamidino-2-phenylindole.2HCl) in 5% (v/v) DMSO, 0.01% (v/v) Tween 20 in PBS for 30 min. All samples were mounted in Citifluor antifade solution (Plano GmbH, Wetzlar, Germany) and viewed with a Leica TCS SP2 AOBS confocal microscope (Leica Microsystems GmbH, Wetzlar, Germany) as described by Takahara *et al*. (2009). Live-cell confocal imaging of infected Arabidopsis epidermal cells expressing RPW8.2–YFP fusion proteins and colocalization with callose was performed as described by Meyer *et al*. (2009). All images were processed using Leica Confocal software (LCS Lite, Leica Microsystems) and Adobe Photoshop 8.0 software.

High-pressure freezing and freeze substitution

High-pressure freezing was performed both on isolated and in planta haustoria. Isolated haustoria suspended in Percoll were collected in cellulose microcapillary tubes (Leica Microsystems GmbH), as described by Hohenberg *et al*. (1994). Tubes were sealed by cutting the ends using a 2-mm-diameter biopsy punch (Stiefel Laboratorium GmbH, Offenbach/Main, Germany). For haustoria in planta, infected leaf tissue was vacuum-infiltrated with MES buffer (20 mM MES, 2 mM $CaCl_2$, 2 mM KCl, 0.2 M sucrose, pH 5.5) twice for 1 min. Discs of tissue were then excised using a biopsy punch. Three to five microcapillary tube segments or one infected leaf disc were placed in aluminium specimen carriers with 100 μm or 300 μm deep cavities, respectively (Leica Microsystems GmbH), mounted in 1-hexadecene, capped with a second specimen carrier (flat side towards the sample) and immediately frozen using a Leica EM HPM 100 high-pressure freezer (Leica Microsystems GmbH).

Freeze substitution was performed in a Leica EM AFS2 freeze substitution device (Leica Microsystems GmbH). For ultrastructural observations, samples were freeze-substituted with anhydrous acetone containing 2% (w/v) osmium tetroxide as follows: 30 h at –85°C, transition to –60°C at 2°C h^{-1} (12.5 h), 30 h at –60°C, transition to –30°C at 2°C h^{-1} (15 h), 30 h at –30°C, transition to –30°C at 2°C h^{-1} (5 h), 18 h at –20°C, transition to 4°C at 24°C h^{-1} (1 h), 1 h at 4°C, 1 h at room temperature. For immunogold labelling, samples were freeze-substituted in acetone containing 0.5 % (w/v) uranyl acetate as follows: 90 h at –85°C, transition to –60°C at 2°C h^{-1} (12.5 h), 24 h at –60°C [samples then transferred to 0.1% (w/v) osmium tetroxide in acetone], transition to –30°C at 2°C h^{-1} (15 h), 24 h at –30°C, transition to –20°C at 1°C h^{-1} (10 h), 12 h at –20°C.

Embedding and sectioning

Samples for ultrastructural observation were rinsed in acetone, embedded in low-viscosity epoxy resin (Science Services, Munich, Germany) for 6 days and polymerized in flat embedding moulds at 60°C for 24 h. Ultrathin (70–90 nm) sections were collected on copper slot grids as described by Moran and Rowley (1987). Sections of 150–200 nm were used for 3D tomographic reconstruction. Samples for immunogold labelling were rinsed in ethanol and embedded in medium-grade LR White resin (Plano GmbH) for 8 days at −20°C. Polymerization with UV light was for 24 h at −20°C and 24 h at 0°C. Ultrathin sections were collected on gold slot grids as above.

Immunogold labelling

For immunogold labelling, sections were blocked for 30 min in a 1:30 dilution of goat normal serum in TRIS buffer (20 mM TRIS, 15 mM NaN_3, 225 mM NaCl, pH 6.9) supplemented with 0.1% (v/v) Tween 20 and 1% (w/v) BSA (TRIS-BSA). After three washes for 10 min in TRIS-BSA, sections were incubated in undiluted primary antibody (see Table 1) at 4°C overnight. Only the anti-β-1,3-glucan antibody was used at a 1:100 dilution. After washing in TRIS-BSA (4 × 10 min), sections were incubated with a 1:20 dilution of the corresponding secondary antibody (goat anti-mouse or goat anti-rat) conjugated to 10 or 20 nm colloidal gold particles (British Biocell International, Cardiff, UK) for 1 h. As negative controls for immunogold labelling, the primary antibody was replaced either with TRIS-BSA or, in case of mouse monoclonal antibodies, with UBIM22 raised to rat bone cells (Mackie *et al*., 1991).

Transmission electron microscopy (TEM)

Sections for ultrastructural observation were either stained with 1% (w/v) potassium permanganate for 1 min or 2% (w/v) uranyl acetate for 10 min, followed by lead citrate for 15 min. Sections for 3D tomography and immunogold labelling were stained with 2% uranyl acetate for 2–5 min followed by lead citrate for 5–10 min. Sections were examined with an Hitachi H-7650 TEM operating at 100 kV fitted with an AMT XR41-M digital camera (Advanced Microscopy Techniques, Danvers, USA). Images were processed using Adobe Photoshop 10.0 software. For 3D tomographic reconstruction, accelerating voltage was set to 120 kV and reconstruction was realized using Hitachi EMIP 05.07 image integrating software.

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Note added in proof

Since our article's submission, a new publication reveals a role of MVBs in the delivery and assembly of callose and other defense components at the papillae formed in barley in response to *B. graminis* f. sp. *hordei* (Böhlenius *et al*., 2010). The authors identified an ADP-ribosylation factor (ARFA1b/1c) that is associated with plant MVBs and is required for penetration resistance mediated by ROR2, the barley homologue of PEN1. ARFA1b/1c normally localises to Golgi and endocytic vesicles and was identified as a component of MVBs, where it may play a role in vesicle budding as well as callose deposition at papillae. This study provides further support for the involvement of non-classical secretion pathways in plant-pathogen interactions on both sides of the biological interface.

References

Aist, J.R. (1976) Papillae and related wound plugs in plant cells. *Annu Rev Phytopathol* **14:** 145–163.

Albuquerque, P.C., Nakayasu, E.S., Rodrigues, M.L., Frases, S., Casadevall, A., Zancope-Oliveira, R.M., *et al*. (2008) Vesicular transport in *Histoplasma capsulatum*: an effective mechanism for *trans*-cell wall transfer of proteins and lipids in ascomycetes. *Cell Microbiol* **10:** 1695–1710.

An, Q., Ehlers, K., Kogel, K.H., von Bel, A.J.E., and Hückelhoven, R. (2006a) Multivesicular compartments proliferate in susceptible and resistant *MLA12*-barley leaves in response to infection by the biotrophic powdery mildew fungus. *New Phytol* **172:** 563–576.

An, Q., Hückelhoven, R., Kogel, K.H., and van Bel, A.J.E. (2006b) Multivesicular bodies participate in a cell wall associated defense response in barley leaves attacked by the pathogenic powdery mildew fungus. *Cell Microbiol* **8:** 1009–1019.

Aziz, A., Gauthier, A., Bezier, A., Poinssot, B., Joubert, J.M., Pugin, A., *et al*. (2007) Elicitor and resistance-inducing activities of β-1,4 cellodextrins in grapevine, comparison with β-1,3 glucans and α-1,4 oligogalacturonides. *J Exp Bot* **58:** 1463–1472.

Bednarek, P., Pislewska-Bednarek, M., Svatos, A., Schneider, B., Doubsky, J., Mansurova, M., *et al*. (2009) A glucosinolate metabolism pathway in living plant cells mediates broad-spectrum antifungal defense. *Science* **323:** 101–106.

Böhlenius, H., Mørch, S.M., Godfrey, D., Nielsen, M.E., and Thordal-Christensen, H. (2010) Multivesicular body-localized GTPase ARFA1b/1c is important for callose deposition and ROR2 syntaxin-dependent preinvasive basal defense in barley. *Plant Cell* doi: 10.1104/tpc.110.078063.

Bracker, C.E. (1968) Ultrastructure of the haustorial apparatus of *Erysiphe graminis* and its relationship to the epidermal cell of barley. *Phytopathology* **58:** 12–30.

Bushnell, W.R. (1972) Physiology of fungal haustoria. *Annu Rev Phytopathol* **10:** 151–176.

Casadevall, A., Nosanchuk, J.D., Williamson, P., and Rodrigues, M.L. (2009) Vesicular transport across the fungal cell wall. *Trends Microbiol* **17:** 158–162.

Catanzariti, A.M., Dodds, P.N., Lawrence, G.J., Ayliffe, M.A., and Ellis, J.G. (2006) Haustorially expressed secreted proteins from flax rust are highly enriched for avirulence elicitors. *Plant Cell* **18:** 243–256.

Celio, G.J., Mims, C.W., and Richardson, E.A. (2004) Ultrastructure and immunocytochemistry of the host–pathogen interface in poinsettia leaves infected with powdery mildew. *Can J Bot* **82:** 421–429.

De Souza Pereira, R., and Geibel, J. (1999) Direct observation of oxidative stress on the cell wall of *Saccharomyces cerevisiae* strains with atomic force microscopy. *Mol Cell Biochem* **201:** 17–24.

Delaney, T.P., Uknes, S., Vernooij, B., Friedrich, L., Weinmann, K., Negrotto, D., *et al*. (1994) A central role of salicylic acid in plant disease resistance. *Science* **266:** 1247–1250.

El Gueddari, N.E., Rauchhaus, U., Moerschbacher, B.M., and Deising, H.B. (2002) Developmentally regulated conversion of surface-exposed chitin to chitosan in cell walls of plant–pathogenic fungi. *New Phytol* **156:** 103–112.

van Esse, H.P., Bolton, M.D., Stergiopoulos, I., de Wit, P.J.G.M., and Thomma, B.P.H.J. (2007) The chitin-binding *Cladosporium fulvum* effector protein Avr4 is a virulence factor. *Mol Plant Microbe Interact* **20:** 1092–1101.

Fujikawa, T., Kuga, Y., Yano, S., Yoshimi, A., Tachiki, T., Abe, K., and Nishimura, M. (2009) Dynamics of cell wall components of *Magnaporthe grisea* during infectious structure development. *Mol Microbiol* **73:** 553–570.

Gil, F., and Gay, J.L. (1977) Ultrastructural and physiological properties of the host interfacial components of haustoria of *Erysiphe pisi in vivo* and *in vitro*. *Physiol Plant Pathol* **10:** 1–12.

Göllner, K., Schweizer, P., Bai, Y., and Panstruga, R. (2008) Natural genetic resources of *Arabidopsis thaliana* reveal a high prevalence and unexpected phenotypic plasticity of *RPW8*-mediated powdery mildew resistance. *New Phytol* **177:** 725–742.

Hajlaoui, M.R., Benhamou, N., and Bélanger, R.R. (1991) Cytochemical aspects of fungal penetration, haustorium formation and interfacial material in rose leaves infected by *Sphaerotheca pannosa* var. *rosae*. *Physiol Mol Plant Pathol* **39:** 341–355.

Hématy, K., Cherk, C., and Somerville, S. (2009) Host–pathogen warfare at the plant cell wall. *Curr Opin Plant Biol* **12:** 406–413.

Hippe, S. (1985) Ultrastructure of haustoria of *Erysiphe graminis* f. sp. *hordei* preserved by freeze-substitution. *Protoplasma* **129:** 52–61.

Hippe-Sanwald, S., Hermanns, M., and Somerville, S. (1992) Ultrastructural comparison of incompatible and compatible interactions in the barley powdery mildew disease. *Protoplasma* **168:** 27–40.

Hohenberg, H., Mannweiler, K., and Müller, M. (1994) High-pressure freezing of cell-suspensions in cellulose capillary tubes. *J Microsc* **175:** 34–43.

Hückelhoven, R. (2005) Powdery mildew susceptibility and biotrophic infection strategies. *FEMS Microbiol Lett* **245:** 9–17.

de Jonge, R., van Esse, H. .P., Kombrink, A., Shinya, T., Desaki, Y., Bours, R., *et al*. (2010) Conserved fungal LysM effector Ecp6 prevents chitin-triggered immunity in plants. *Science* **329:** 953–955.

Knox, J.P., and Roberts, K. (1989) Carbohydrate antigens and lectin receptors of the plasma membrane of carrot cells. *Protoplasma* **152:** 123–129.

Knox, J.P., Linstead, P.J., King, J., Cooper, C., and Roberts, K. (1990) Pectin esterification is spatially regulated both within cell walls and between developing tissues of root apices. *Planta* **181:** 512–521.

Koh, S., Andre, A., Edwards, H., Ehrhardt, S., and Somerville, S. (2005) *Arabidopsis thaliana* subcellular responses to compatible *Erysiphe cichoracearum* infections. *Plant J* **44:** 516–529.

Leckie, C.P., Callow, J.A., and Green, J.R. (1995) Reorganisation of the endoplasmic reticulum in pea leaf epidermal cells infected by the powdery mildew fungus *Erysiphe pisi*. *New Phytol* **131:** 211–221.

Littlefield, L.J., and Bracker, C.E. (1972) Ultrastructural specialization at the host–pathogen interface in rust-infected flax. *Protoplasma* **74:** 271–305.

Mackie, A.J., Roberts, A.M., Callow, A.J., and Green, J.R. (1991) Molecular differentiation in pea powdery-mildew: identification of a 62-kDa N-linked glycoprotein unique to the haustorial plasma membrane. *Planta* **183:** 399–408.

Meikle, P.J., Bonig, I., Hoogenraad, N.J., Clarke, A.E., and Stone, B.A. (1991) The location of 1,3-β-glucans in the walls of pollen tubes of *Nicotiana alata* using a 1, 3-β-glucan-specific monoclonal antibody. *Planta* **185:** 1–8.

Meyer, D., Pajonk, S., Micali, C., O'Connell, R., and Schulze-Lefert, P. (2009) Extracellular transport and integration of plant secretory proteins into pathogen-induced cell wall compartments. *Plant J* **57:** 986–999.

Micali, C., Göllner, K., Humphry, M., Consonni, C., and Panstruga, R. (2008) The powdery mildew disease of Arabidopsis: a paradigm for the interaction between plants and biotrophic fungi. In *The Arabidopsis Book*. Rockville, MD: American Society of Plant Biologists. doi:10.1199/tab.0115 [WWW document]. http://www.aspb.org/publications/arabidopsis/

Mitchell, A., Hutchison, K.A., Pain, N.A., Callow, J.A., and Green, J.R. (1997) A monoclonal antibody that recognises a carbohydrate epitope on N-linked glycoproteins is restricted to a subset of chitin-rich fungi. *Mycol Res* **101:** 73–79.

Miya, A., Albert, P., Shinya, T., Desaki, Y., Ichimura, K., Shirasu, K., *et al*. (2007) CERK1, a LysM receptor kinase is essential for chitin elicitor signalling in Arabidopsis. *Proc Nat Acad Sci USA* **104:** 19613–19618.

Moran, D.T., and Rowley, J.C. (1987) Biological specimen preparation for correlative light and electron microscopy. In *Correlative Microscopy in Biology: Instrumentation and Methods*. Hayat, M. A. (ed.). New York: Academic Press, pp. 1–22.

O'Connell, R.J., and Panstruga, R. (2006) Tête à tête inside a plant cell: establishing compatibility between plants and biotrophic fungi and oomycetes. *New Phytol* **171:** 699–718.

Pain, N.A., Green, J.R., Gammie, F., and O'Connell, R.J. (1994) Immunomagnetic isolation of viable intracellular hyphae of *Colletotrichum lindemuthianum* (Sacc. & Magn.) Briosi & Cav. from infected bean leaves using a monoclonal antibody. *New Phytol* **127:** 223–232.

Puhlmann, J., Bucheli, E., Swain, M.J., Dunning, N., Abersheim, P., Darvill, A.G., and Hahn, M.G. (1994) Generation of monoclonal antibodies against plant cell wall polysaccharides. I. Characterisation of a monoclonal antibody to a terminal alpha-(1,2)-linked fucosyl-containing epitope. *Plant Physiol* **104:** 699–710.

Ramonell, K., Berrocal-Lobo, M., Koh, S., Wan, J., Edwards, H., Stacey, G., and Somerville, S. (2005) Loss-of-function mutations in chitin responsive genes show increased susceptibility to the powdery mildew pathogen *Erysiphe cichoracearum*. *Plant Physiol* **138:** 1027–1036.

Rappleye, C.A., Groppe Eissenberg, L., and Goldman, W.E. (2007) *Histoplasma capsulatum* α-(1,3)-glucan blocks innate immune recognition by the β-glucan receptor. *Proc Nat Acad Sci USA* **104:** 1366–1370.

Reuber, T.L., Plotnikova, J.M., Dewdney, J., Rogers, E.E., Wood, W., and Ausubel, F.M. (1998) Correlation of defense gene induction defects with powdery mildew susceptibility in Arabidopsis enhanced disease susceptibility mutants. *Plant J* **16:** 473–485.

Ridout, C., Skamnioti, P., Porrit, O., Sacristan, S., Jones, J.D., and Brown, J.K. (2006) Multiple avirulence paralogues in cereal powdery mildew may contribute to parasite fitness and defeat of plant resistance. *Plant Cell* **18:** 2402–2414.

Roberts, A.M., Mackie, A.J., Hathaway, V., Callow, J.A., and Green, J.R. (1993) Molecular differentiation in the extrahaustorial membrane of pea powdery mildew haustoria at early and late stages of development. *Physiol Mol Plant Pathol* **43:** 147–160.

Rodrigues, M.L., Nakayasu, E.S., Oliveira, D.L., Nimrichter, L., Nosanchuk, J.D., Almeida, I.C., and Casadevall, A. (2008a) Extracellular vesicles produced by *Cryptococcus neoformans* contain protein components associated with virulence. *Eukaryot Cell* **7:** 58–67.

Rodrigues, M.L., Nimrichter, L., Oliveira, D.L., Nosanchuk, J.D., and Casadevall, A. (2008b) Vesicular *trans*-cell wall transport in fungi: a mechanism for the delivery of virulence-associated macromolecules? *Lipid Insights* **2:** 27–40.

Schorey, J.S., and Bhatnagar, S. (2008) Exosome function: from tumor immunology to pathogen biology. *Traffic* **9:** 871–881.

Schwessinger, B., and Zipfel, C. (2008) News from the frontline: recent insights into PAMP-triggered immunity in plants. *Curr Opin Plant Biol* **11:** 389–395.

Skalamera, D., Jibodh, S., and Heath, M. (1997) Callose deposition during the interaction between cowpea (*Vigna unguiculata*) and the monokaryotic stage of the cowpea rust fungus (*Uromyces vignae*). *New Phytol* **136:** 511–524

Smallwood, M., Beven, A., Donovan, N., Neill, S.J., Peart, J., Roberts, K., and Knox, J.P. (1994) Localisation of cell wall proteins in relation to the developmental anatomy of the carrot root apex. *Plant J* **5:** 237–246.

Stark-Urnau, M., and Mendgen, K. (1995) Sequential deposition of plant glycoproteins and polysaccharides at the host–parasite interface of *Uromyces vignae* and *Vigna sinensis* – evidence for endocytosis and secretion. *Protoplasma* **186:** 1–11.

Takahara, H., Dolf, A., Endl, E., and O'Connell, R. (2009) Flow cytometric purification of *Colletotrichum higginsianum* biotrophic hyphae from Arabidopsis leaves for stage specific transcriptome analysis. *Plant J* **59:** 672–683.

Takamatsu, S. (2004) Phylogeny and evolution of the powdery mildew fungi (Erysiphales, Ascomycota) inferred from nuclear ribosomal DNA sequences. *Mycoscience* **45:** 147–157.

Thordal-Christensen, H., Zhang, Z.G., Wei, Y.D., and Collinge, D.B. (1997) Subcellular localization of H_2O_2 accumulation in papillae and hypersensitive response during the barley–powdery mildew interaction. *Plant J* **11:** 1187–1194.

Voegele, R.T., and Mendgen, K. (2003) Rust haustoria: nutrient uptake and beyond. *New Phytol* **159:** 93–100.

Wan, J.R., Zhang, X.C., Neece, D., Ramonell, K.M., Clough, S., Kim, S.Y., *et al.* (2008) A LysM receptor-like kinase plays a critical role in chitin signaling and fungal resistance in Arabidopsis. *Plant Cell* **20:** 471–481.

Wang, W., Wen, Y., Berkey, R., and Xiao, S. (2009) Specific targeting of the Arabidopsis resistance protein RPW8.2 to the interfacial membrane encasing the fungal haustorium renders broad-spectrum resistance to powdery mildew. *Plant Cell* **21:** 2898–2913.

Wheeler, A.M., and Hartley, M.R. (1975) Major mRNA species from spinach chloroplasts do not contain poly(A). *Nature* **257:** 66–67.

Yun, B., Atkinson, H.A., Gaborit, C., Greenland, A., Read, N.D., Pallas, J. .A., and Loake, G.J. (2003) Loss of actin cytoskeletal function and EDS1 activity, in combination, severely compromises non-host resistance in Arabidopsis against wheat powdery mildew. *Plant J* **34:** 768–777.

Zeyen, R.J., and Bushnell, W.R. (1979) Papilla response of barley epidermal cells caused by *Erysiphe graminis*: rate and method of deposition determined by microcinematography and transmission electron microscopy. *Can J Bot* **58:** 898–913.

Zeyen, R.J., Carver, T.L.W., and Lyngkjaer, M.F. (2002) Epidermal cell papillae. In *The Powdery Mildews: A Comprehensive Treatise*. Bélanger, R.R., Bushnell, W.R., Dik, A.J., and Carver, T.L.W. (eds). St. Paul, MN: The American Phytopathological Society, pp. 107–125.

Supporting information

Additional Supporting Information may be found in the online version of this article:

Fig. S1. Light micrographs showing the development of *G. orontii* infection structures on *A. thaliana NahG* cotyledons.

Fig. S2. Electron tomography of the haustorial interface of *G. orontii* infecting an *A. thaliana* epidermal cell. (a) TEM orientation image. (b) Animated video of a 3D tomogram showing the same area as in (a).

Fig. S3. Ultrastructure of the plant endoplasmic reticulum (ER) around *G. orontii* haustoria.

Fig. S4. Immunogold localization of plant cell wall components at the plant cell-haustorium interface in *A. thaliana NahG* epidermal cells infected by *G. orontii* haustoria.

Fig. S5. Immunogold labelling of fungal cell wall polymers in *G. orontii* haustorial cell walls *in planta*.